Individual Diagnostic Requirement Specification

CN200 ECM (1.8L)

Version: 1.0

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Requirement wording conventions

Within this document the following conventions are applied when describing specific compliance requirements:

The word “Should” denotes a preference or desired conformance.

The word “Shall” is used in the following ways:

1. To state a binding requirement on the special network, or the ECUs which comprise the network, which is verifiable by external manipulation and/or observation of an ECU or the network.

2. To state a binding requirement upon an ECU’s requirements document that is verifiable through a review of the document.

The word “Must” is used to state a binding requirement upon ECUs on the network which have a corresponding ECU requirements document. These requirements will be verified as part of the component verification.

NOTE: In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence.

NOTE: Nothing in this specification supersedes applicable laws and regulations unless a specific exemption has been obtained.

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History record of the document

The following table contains the revision history for this specification:

Version#	Date	Editor	Description of the revision
V1.0	2013-03-20	Liao wenqing	Initial release

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1 Introduction

1.1 Scope

The detailed description about this chapter can be found in chapter 1.1 of the document <CN200 General Diagnostic Requirement Specification>.

NOTE: For CN200 vehicle, ECM need to realize the software download function, the detailed requirements about software download can be found in chapter 16 of the document <CN200 General Diagnostic Requirement Specification>.

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2 Normative references

This document references ISO, GM, SGMW and other publications as listed in the following sub sections.

2.1 ISO Publications

[ISO 11898-2]	Road Vehicles - Controller Area Networks (CAN) - Part 2: High-speed medium access unit ISO 2003
[ISO 15765-1]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information ISO 2004
[ISO 15765-2]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services ISO 2004
[ISO 15765-3]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of diagnostic services ISO 2004
[ISO 14229]	Road Vehicles - Unified diagnostic services (UDS) - Specification and requirements ISO 2006
[ISO 15031-5]	Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 5: Emission-related diagnostic services ISO 2005
[ISO 15031-6]	Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 6: Diagnostic trouble code definitions ISO 2005

2.2 GM Publications

GMW 3107	GMLAN Functional Handler Specification
GMW 3104	GMLAN Communication Strategy Specification
GMW 3110	GMLAN Enhanced Diagnostic Test Mode Specification
GMW 3122	GM High Speed CAN Physical Layer Specification

2.3 SGMW Standard

CN200 General Diagnostic Requirement Specification

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3 Terms, definitions and abbreviations

3.1 ISO Terms

This document makes use of terms defined in the ISO 15765-2, ISO 15765-3 and ISO 14229-1.

3.2 Specific Terms Defined By This Document

Table 3-1Specific Terms Defined By This Document

Abbreviation	Description
ECM	**E**ngine **C**ontrol **M**odule
ABS	**A**nti-lock **B**rake **S**ystem
BCM	**B**ody **C**ontrol **M**odule
EPS	**E**lectronic **P**ower **S**teering
IC	**I**nstrument **C**luster
SDM	Restraint **S**ensing and **D**iagnostic **M**odule
TCM	**T**ransmission **C**ontrol **M**odule

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4 General requirement

All the general requirements, with which the tester and ECM shall comply, can be found in section 4 of the document <CN200 General Diagnostic Requirement Specification>.
It is possible for ECM supplier to add the additional definitions for general requirement in this section. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

4.1 Diagnostic CAN identifiers definition

The following table provides an overview about the definition of ECM diagnostic CAN identifier.

Table 4-1 ECM diagnostic CAN identifier

ECU		Functionally addressed request from tester to ECUs	Physically addressed request from tester to ECU	Response from ECU to tester
Name	Emission-related OBDECU (YES/NO)			
ECM	YES	0x7DF	0x7E0	0x7E8

4.2 ECU Diagnostic Address definition

The following table provides an overview about the definition of ECM diagnostic Address.

Table 4-2 ECM Diagnostic Address

ECU Name	Description	Value	Comments
ECM	ECUDiagnosticAddress	0x11	ECM diagnostic address, which is defined in CN200 General Diagnostic Requirement Specification.

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5 Physical layer requirement

All the physical layer requirements, with which the tester and ECM shall comply, can be found in section 5 of the document <CN200 General Diagnostic Requirement Specification>.
It is possible for ECM supplier to add the additional definitions for physical layer requirement in this section. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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6 Data link layer requirement

All the data link layer requirements, with which the tester and ECM shall comply, can be found in section 6 of the document <CN200 General Diagnostic Requirement Specification>.
It is possible for ECM supplier to add the additional definitions for data link layer requirement in this section. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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7 Network layer requirement

All the network layer requirements, with which the tester and ECM shall comply, can be found in section 7 of the document <CN200 General Diagnostic Requirement Specification>.
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8 Application and session layers requirement

All the application and session layer requirements, with which the tester and ECM shall comply, can be found in section 8 of the document <CN200 General Diagnostic Requirement Specification>.
It is possible for ECM supplier to add the additional definitions for application and session layer requirement in this section. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

8.1 Diagnostic Service Table Overview

The table followed (Table 8-1) shows the services which are allowed during the defaultSession, programmingSession and extendedSession of ECM. ECM is allowed to support more other diagnostic services per session after confirmed by SGMW.

Table 8-1 Diagnostic Service Table Overview

Service Name	SID (Hex)	Default Session	programming Session	Extended Session	Supported Addressed
Diagnostic and communication management functional unit					
DiagnosticSessionControl	10	×	×	×	P&F
ECUReset	11	×	×	×	P&F
SecurityAccess	27	-	×	×	P
CommunicationControl	28	-	-	×	P&F
TesterPresent	3E	×	×	×	P&F
AccessTimingParameter	83	-	-	-	-
SecuredDataTransmission	84	-	-	-	-
ControlDTCSetting	85	-	-	×	P&F
ResponseOnEvent	86	-	-	-	-
LinkControl	87	-	-	-	-
Data transmission functional unit					
ReadDataByIdentifier	22	×	×	×	P&F
ReadMemoryByAddress	23	-	-	-	-
ReadScalingDataByIdentifier	24	-	-	-	-
ReadDataByPeriodicIdentifier	2A	-	-	-	-
DynamicallyDefineDataIdentifier	2C	×	-	×	P
WriteDataByIdentifier	2E	-	×★	×★	P
WriteMemoryByAddress	3D	-	-	-	-
Stored data transmission functional unit					
ClearDiagnosticInformation	14	×	-	×	P&F
ReadDTCInformation	19	×	-	×	P&F
InputOutput control functional unit					
InputOutputControlByidentifier	2F	×	-	×☆	P

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Remote activation of routine functional unit					
RoutineControl	31	-	×★	×☆	P
Upload download functional unit					
RequestDownload	34	-	×★	-	P
RequestUpload	35	-	-	-	-
TransferData	36	-	×	-	P
RequestTransferExit	37	-	×	-	P

1. “×”: support without security access; “×☆”: support, whether or not need security access depend on the used DID or routineidentifier; “×★”: support with security access; “-”: NOT supported.
2. “P”: physically; “F”: Functionally.

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9 Diagnostic service documentation structure

All the requirements about diagnostic service documentation structure, with which the tester and ECM shall comply, can be found in section 9 of the document <CN200 General Diagnostic Requirement Specification>.

It is possible for ECM supplier to add the additional definitions for diagnostic service documentation structure in this section. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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10 Diagnostic and communication management functional unit

10.1 DiagnosticSessionControl (10 hex)

10.1.1 Request message

Table 10-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Request Service Id	10
#2	DiagnosticSessionType=[DefaultSession ProgrammingSession ExtendedDiagnosticSession]	 01 02 03

10.1.2 Positive response message

Table 10-2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Response Service Id	50
#2	DiagnosticSessionType=[DefaultSession ProgrammingSession ExtendedDiagnosticSession]	 01 02 03
 #3 #4 #5 #6	sessionParameterRecord[]#1=[$P2_{CAN_ECU_max}$(high byte) $P2_{CAN_ECU_max}$(low byte) $P2^*_{CAN_ECU_max}$(high byte) $P2^*_{CAN_ECU_max}$(low byte)]	 00 32 01 F4

10.1.3 Supported negative response codes

Table 10-3 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

10.2 ECUReset (11 hex)

10.2.1 Request message

Table 10-4 Request message definition

A_Data Byte	Parameter name	Hex value
#1	ECUReset Request Service Id	11
#2	resetType =[hardReset softReset]	 01 03

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10.2.2 Positive response message

Table 10-5 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ECUReset Response Id	51
#2	resetType =[hardReset softReset]	 01 03

10.2.3 Supported negative response codes

Table 10-6 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ECM to implement the request message shall be defined by the ECM supplier.*

10.3 SecurityAccess (27 hex)

10.4 CommunicationControl (28 hex)

10.5 TesterPresent (3E hex)

10.5.1 Request message

Table 10-7 Request message definition

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Request Service Id	3E
#2	zeroSubFunction=[zeroSubFunction (suppressPosRspMsgIndicationBit =0) zeroSubFunction (suppressPosRspMsgIndicationBit =1)]	 00 80

10.5.2 Positive response message

Table 10-8 Positive response message definition(sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =0))

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Response Service Id	7E
#2	zeroSubFunction= zeroSubFunction (suppressPosRspMsgIndicationBit =0)	00

sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of

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suppressPosRspMsgIndicationBit =1.

10.5.3 Supported negative response codes

Table 10-9 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

10.6 AccessTimingParameter (83 hex)

N/A

10.7 SecuredDataTransmission (84 hex)

N/A

10.8 ControlDTCSetting (85 hex)

10.9 ResponseOnEvent (86 hex)

N/A

10.10 LinkControl (87 hex)

N/A

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11 Data transmission functional unit

11.1 ReadDataByIdentifier (22 hex)

NOTE: The number of dataIdentifier that can be simultaneously requested should be determined by SGMW and ECU Supplier together.

NOTE: The DID which used to identify the general data record for all the ECUs called vehicle-manufacturer-specific DID, the definition about vehicle-manufacturer-specific DID can be found in chapter C.1 of Annex C. The DID which used to identify the individual data record for each ECU called ECU-supplier-specific DID, the definition about ECU-supplier-specific DID can be found in in chapter C.2 of Annex C.

11.1.1 Request message

Table 11-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Request Service Id	22
#2 #3	dataIdentifier[]#1=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
:	:	:
#n-1 #n	dataIdentifier[]#m[a]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
a. m≤number (This number should be determined by SGMW and ECU Supplier together)		

11.1.2 Positive response message

Table 11-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Response Service Id	62
#2 #3	dataIdentifier[]#1=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4 : #(k-1)+4	dataRecord[]#1=[data #1 : data #k [a]]	00-FF : 00-FF
:	:	:
#n-(o-1)-2 #n-(o-1)-1	dataIdentifier[]#m[b]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#n-(o-1) : #n	dataRecord[]#m[b]=[data #1 : data #o [a]]	00-FF : 00-FF

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a. The number of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C. b. m≤number (This number should be determined by SGMW and ECU Supplier together)

11.1.3 Supported negative response codes

Table 11-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ECM to implement the request message shall be defined by the ECM supplier.*

11.2 ReadMemoryByAddress (23 hex)

N/A

11.3 ReadScalingDataByIdentifier (24 hex)

N/A

11.4 ReadDataByPeriodicIdentifier (2A hex)

N/A

11.5 DynamicallyDefineDataIdentifier (2C hex)

11.6 WriteDataByIdentifier (2E hex)

11.6.1 Request message

Table 11-4 Request message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Request Service Id	2E
#2 #3	dataIdentifier[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4 : #m+3	dataRecord[]=[data #1 : data #m[a]]	00-FF : 00-FF
a. The content of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C.		

11.6.2 Positive response message

Table 11-5 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Response Service Id	6E
#2 #3	dataIdentifier[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF

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11.6.3 Supported negative response codes

Table 11-6 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
33	**securityAccessDenied**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ECM to implement the request message shall be defined by the ECM supplier.*
72	**generalProgrammingFailure**

11.7 WriteMemoryByAddress (3D hex)

N/A

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12 Stored data transmission functional unit

12.1 ClearDiagnosticInformation (14 hex)

12.1.1 Request message

Table 12-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Request Service Id	14
#2 #3 #4	groupOfDTC[]=[groupOfDTCHighByte groupOfDTCMiddleByte groupOfDTCLowByte]	 FF FF FF

12.1.2 Positive response message

Table 12-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Response Service Id	54

12.1.3 Supported negative response codes

Table 12-3 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ECM to implement the request message shall be defined by the ECM supplier.*

12.2 ReadDTCInformation (19 hex)

NOTE: ECM supplier should provide SGMW with DTC list which ECM supported.

NOTE: The maximum number of DTC that ECM can store simultaneously should be determined by SGMW and ECU Supplier together.

NOTE: If a DTC only supports a group snapshot data record, the maximum number of DTC snapshot record that ECM can store simultaneously should the same as the maximum number of DTC.

NOTE: From an attribute point of view, snapshot is divided into two parts: one is the general snapshot with one or more group data records for all the ECUs; Tthe other is local snapshot with one or more group data records for each specified ECU (ECM). The format and content about snapshot should be determined by SGMW and ECM Supplier together.

NOTE: If all fault memories are occupied, while the new DTC is generated, the Overflow strategy about stored DTCs and snapshot data records should be determined by SGMW and ECM supplier

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together.

12.2.1 Request message

Table 12-4 Request message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=[reportNumberOfDTCByStatusMask reportDTCByStatusMask]	 01 02
#3	DTCStatusMask	00-FF

12.2.2 Positive response message

Table 12-5 Positive response message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportNumberOfDTCByStatusMask	01
#3	DTCStatusAvailabilityMask	00-FF
#4	ISO 15031-6 DTCFormat	00
 #5 #6	DTCCount[]=[DTCCountHighByte DTCCountLowByte]	 00-FF 00-FF

Table 12-6 Positive response message definition (sub-function = reportDTCByStatusMask, reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=[reportDTCByStatusMask	02
#3	DTCStatusAvailabilityMask	00-FF
 #4 #5 #6 #7 #8 #9 #10 #11 : #n-3 #n-2 #n-1 #n	DTCAndStatusRecord[] = [DTCHighByte #1 DTCMiddleByte #1 DTCLowByte #1 statusOfDTC #1 DTCHighByte #2 DTCMiddleByte #2 DTCLowByte #2 statusOfDTC #2 : DTCHighByte #m DTCMiddleByte #m DTCLowByte #m statusOfDTC #m]	 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF : 00-FF 00-FF 00-FF 00-FF

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13 InputOutput control functional unit

13.1 InputOutputControlByIdentifier (2F hex)

13.1.1 Request message

Table 13-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	InputOutputControlByIdentifier Request Service Id	2F
#2 #3	dataIdentifier#1[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4	inputOutputControlParameter=returnControlToECU	00

Table 13-2 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	InputOutputControlByIdentifier Request Service Id	2F
#2 #3	dataIdentifier#1[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4	inputOutputControlParameter=shortTermAdjustment	03
#5 : #5+(m-1)	controlState#1 : controlState#m[a]	00-FF : 00-FF
a. The number of controlState# depends on specified dataIdentifier in chapter E.2 of Annex E.		

13.1.2 Positive response message

Table 13-3 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	InputOutputControlByIdentifier Response Service ID	6F
#2 #3	dataIdentifier#1[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4	inputOutputControlParameter=returnControlToECU	00

Table 13-4 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	InputOutputControlByIdentifier Response Service ID	6F
#2 #3	dataIdentifier#1[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4	inputOutputControlParameter=shortTermAdjustment	03
#5 : #5+(m-1)	controlState#1 : controlState#m[a]	00-FF : 00-FF
a. The number of controlState# depends on specified dataIdentifier in chapter E.2 of Annex E.		

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13.1.3 Supported negative response codes

Table 13-5 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
33	**securityAccessDenied**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ECM to implement the request message shall be defined by the ECM supplier.*

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14 Remote activation of routine functional unit

14.1 RoutineControl (31 hex)

14.1.1 Request message

Table 14–1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	RoutineControl Request Service Id	31
#2	RoutineControlType=[startRoutine stopRoutine requestRoutineResults]	 01 02 03
 #3 #4	routineIdentifier[]=[byte#1(MSB) byte#2(LSB)]	 00-FF 00-FF
 #5 : #m+5	routineControlOptionRecord[]=[routineControlOption#1 : routineControlOption#m[a]]	 00-FF : 00-FF

a. The number of routineControlOption# depends on specified routineIdentifier in chapter F.2 of Annex F.

14.1.2 Positive response message

Table 14–2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	RoutineControl Response Service Id	71
#2	RoutineControlType=[startRoutine stopRoutine requestRoutineResults]	 01 02 03
 #3 #4	routineIdentifier[]=[byte#1 (MSB) byte#2 (LSB)]	 00-FF 00-FF
 #5 : #m+5	routineStatusRecord[]=[routineStatus #1 : routineStatus #m[a]]	 00-FF : 00-FF

a. The number of routineControlOption# depends on specified routineIdentifier in chapter F.2 of Annex F.

14.1.3 Supported negative response codes

Table 14–3 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
33	**securityAccessDenied**
12	**subFunctionNotSupported**
31	**requestOutOfRange**

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13	**incorrectMessageLengthOrInvalidFormat**
24	**requestSequenceError**
22	**conditionsNotCorrect** This code shall be returned if the criteria for the request InputOutputControl are not met. *The corresponding correct condition for ECM to implement the request message shall be defined by the ECM supplier.*

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15 Upload download functional unit

15.1 RequestDownload (34 hex)

15.2 RequestUpload (35 hex)

N/A

15.3 TransferData (36 hex)

N/A

15.4 RequestTransferExit (37 hex)

N/A

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16 ECU programming requirements and process

The detailed description about this chapter can be found in chapter 16 of the document <CN200 General Diagnostic Requirement Specification>.

NOTE: For CN200 vehicle, ECM need to realize the software download function, the detailed requirements about software download can be found in chapter 16 of the document <CN200 General Diagnostic Requirement Specification>.

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Annex A (informative) Global parameter definitions

The detailed description about this Annex can be found in Annex A of the document <CN200 General Diagnostic Requirement Specification>.

It is possible for ECM supplier to add the additional definitions in Annex A. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex B (normative) Diagnostic and communication management functional unit data parameter definitions

The detailed description about this Annex can be found in Annex B of the document <CN200 General Diagnostic Requirement Specification>.

It is possible for ECM supplier to add the additional definitions in Annex B. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex C (normative) Data transmission functional unit data parameter definitions

C.1 Vehicle-manufacturer-specific DID definitions

The following table lists the definition about vehicle-manufacturer-specific DID.

DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
		22	2E		
F190	SGMW VIN	x	x	17	ASCII
NOTE: "×": support; "-": not supported					

C.1.12 DID 0xF190-SGMW VIN

Byte No.	Description	Data Type	Value (Hex)
data #1	VIN(high byte)	ASCII	the real number
:	:	:	:
data #17	VIN(low byte)	ASCII	the real number
NOTE: The default value is 0x20 in ASCII.			

C.2 ECU-supplier-specific DID definitions

The following table lists the definition about ECU-supplier-specific DID. According to the requirements of the document <CN200 General Diagnostic Requirement Specification>, the range which ECM used is 0x0101~0x0FFF.

DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
		22	2E		
0200	maximum attainable indicated torque	x	-	2	Hex
0201	desired indicated engine torque	x	-	2	Hex
0203	engine temperature	x	-	1	Hex
0204	engine speed	x	-	2	Hex
0205	time counter at end of start	x	-	2	Hex
0206	fuel level	x	-	2	Hex
0209	ambient pressure	x	-	2	Hex
020B	throttle angle with respect to lower mechanical stop	x	-	2	Hex
020D	intake air temperature	x	-	1	Hex
020E	ambient air temperature	x	-	1	Hex
020F	pressure upstream throttle valve	x	-	2	Hex
0210	normalized angle acceleration pedal	x	-	2	Hex
0211	volumetric flow of fuel in L/s	x	-	2	Hex

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DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
0212	desired pressure upstream throttle valve	x	-	2	Hex
0214	exhaust gas temperature at manifold out of model	x	-	2	Hex
021B	air-mass flow HFM	x	-	2	Hex
021C	ISO15031_PID03a	x	-	1	Hex
021D	ISO15031_PID03b	x	-	1	Hex
021E	relative air mass (calc. load value) acc. To SAE J1979	x	-	1	Hex
0220	Air mass flow	x	-	1	Hex
0223	output voltage oxygen sensor downstream catalyst	x	-	1	Hex
0224	output voltage oxygen sensor downstream catalyst bank 2	x	-	1	Hex
0225	relative air charge	x	-	1	Hex
0227	ETS-path as environmental condition for diagnosis of function monitoring	x	-	1	Hex
0228	Torque path in function and function monitoring as environmental cond. of diag	x	-	1	Hex
0229	actual torque at response of torque comparison in the function monitoring	x	-	1	Hex
022B	model-based substitute value for engine temperature signal in case of error	x	-	1	Hex
022C	desired throttle angle w.r.t. to lower mechanical stop	x	-	1	Hex
022D	exhaust temperature downstream catalyst (model)	x	-	1	Hex
022F	Engine coolant temperature, linearised and calculated	x	-	1	Hex
0230	model-based reference value for plausibility check of engine temperature signal	x	-	2	Hex
0231	Calculated actual torque in function monitoring	x	-	1	Hex
0232	Engaged gear	x	-	1	Hex
0234	relative fuel mass for storage (service tool)	x	-	2	Hex
0236	real ignition angle	x	-	1	Hex
0237	correction factor for mass-flow substitute load signal	x	-	2	Hex
023A	throttle blade position dependent on poti 1	x	-	2	Hex
023B	throttle blade position dependent on poti 2	x	-	2	Hex
023C	DLR for DV-E: Sum of the PID-components (Throttle actuator Control PWM)	x	-	2	Hex
023D	Diagnosis: 8 bit copy of wdknlp_w as ambient value	x	-	1	Hex
023F	Voltage DK-Poti 1 at the NLP	x	-	2	Hex
0240	sensor voltage poti 1 of throttle actuator at (lower) mechanical stop	x	-	1	Hex
0241	Automobile acceleration in the longitudinal direction	x	-	1	Hex

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DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
0243	signal voltage, ambient pressure sensor	x	-	2	Hex
0244	Ambient pressure without substitution on sensor error	x	-	2	Hex
0245	intake manifold pressure (absolute)	x	-	2	Hex
0246	Number of clutch operations detected	x	-	1	Hex
0247	intake air temperature, linearised and calculated	x	-	1	Hex
0248	vehicle speed output value to scan tool	x	-	2	Hex
0249	battery voltage; scanned value of ADC	x	-	2	Hex
0252	Angle: intake valve opens related to LWOT	x	-	2	Hex
0254	desired camshaft angle inlet valve opened	x	-	2	Hex
0256	oil temperature	x	-	2	Hex
0257	lambda controller output	x	-	2	Hex
0258	multiplicative correction of the mixture adaptation	x	-	2	Hex
0259	Lambda controller output bank 2	x	-	2	Hex
025A	multiplicative correction of the mixture adaptation bank 2	x	-	2	Hex
025B	sensor voltage from throttle potentiometer 1	x	-	2	Hex
025C	sensor voltage from throttle potentiometer 2	x	-	2	Hex
0260	additive correction of the mixture adaptation	x	-	2	Hex
0262	Voltage PWG potentiometer 1	x	-	2	Hex
0263	Voltage PWG potentiometer 2	x	-	2	Hex
0264	Doubled PWG potentiometer-2 voltage	x	-	2	Hex
0265	Actual value internal res. Ri-Nernst cell lambda sensor downstream cat	x	-	2	Hex
0266	Exhaust gas temperature in front of precatalyst out of model	x	-	2	Hex
0267	HFM air mass value	x	-	2	Hex
0268	Transfer function (Mrad/Mkurbelwelle) from transmission control	x	-	1	Hex
026B	Interface for mode $01+$02 short term fuel trim bank 1 for PID $06	x	-	1	Hex
026C	Interface for mode $01+$02 long term fuel trim bank 1 for PID $07	x	-	1	Hex
026D	injection time	x	-	4	Hex
026E	effective steady-state idle speed	x	-	1	Hex
0278	Location of oxygen sensors	x	-	1	Hex
027A	OBD requirements to which vehicle is designed	x	-	1	Hex
027B	counter for distance travelled while MIL is activated	x	-	2	Hex
027C	engine temperature at start	x	-	1	Hex
027D	standardized mass flow from canister purge control	x	-	2	Hex
027E	basic ignition angle	x	-	1	Hex
027F	fault counter, summary, counts emission relevant misfirings of all cylinders	x	-	2	Hex
0284	area of misfire, minimum enginge speed	x	-	2	Hex

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DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
0285	area of misfire, maximum engine speed	x	-	2	Hex
0286	area of misfirings, minimum load	x	-	1	Hex
0287	area of misfirings, maximum load	x	-	1	Hex
0288	Cycle time forced lambda modulation	x	-	2	Hex
0289	desired torque change from the idle speed control (PD-part)	x	-	2	Hex
02A2	Actual intake manifold pressure sensor voltage	x	-	2	Hex
02AA	O_2 sensor voltage bank1, sensor 1	x	-	2	Hex
02AB	O_2 sensor voltage bank1, sensor 2	x	-	2	Hex
0311	ignition-timing retardation with retardation for dynamics, Cylinder #1	x	-	1	Hex
0312	ignition-timing retardation with retardation for dynamics, Cylinder #2	x	-	1	Hex
0313	ignition-timing retardation with retardation for dynamics, Cylinder #3	x	-	1	Hex
0314	ignition-timing retardation with retardation for dynamics, Cylinder #4	x	-	1	Hex
0321	reference level knock control, 16bit, Cylinder #1	x	-	2	Hex
0322	reference level knock control, 16bit, Cylinder #2	x	-	2	Hex
0323	reference level knock control, 16bit, Cylinder #3	x	-	2	Hex
0324	reference level knock control, 16bit, Cylinder #4	x	-	2	Hex
0331	number of misfire: cylinder#1	x	-	2	Hex
0332	number of misfire: cylinder#2	x	-	2	Hex
0333	number of misfire: cylinder#3	x	-	2	Hex
0334	number of misfire: cylinder#4	x	-	2	Hex
0335	Cycle time forced lambda modulation	x	-	2	Hex
0D00	battery voltage	x	-	2	Hex
0D01	vehicle speed	x	-	2	Hex
0D05	indicated real engine torque	x	-	2	Hex
D106	Drivingdistance since prime start	x	-	4	Hex
NOTE: "×": support; "-": not supported					

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Annex D (normative) Stored data transmission functional unit data parameter definitions

The detailed description about this Annex can be found in Annex D of the document <CN200 General Diagnostic Requirement Specification>.

It is possible for ECM supplier to add the additional definitions in Annex D. For CN200 vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex E (normative) Input output control functional unit data parameter definitions

E.1 InputOutputControlParameter definitions

E.2 InputOutput ControlState definitions

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Annex F (normative)Remote activation of routine functional unit data parameter definitions

F.1 RoutineIdentifier definition

F.2 RoutineControl routineStatus definitions

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Annex G (informative) Examples for addressAndLengthFormatIdentifier parameter values

The detailed description about this Annex can be found in Annex G of the document <CN200 General Diagnostic Requirement Specification>.

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